

$$Y = f(X)$$

$$Y = W_0 + W_1 \cdot X_1 + W_2 \cdot X_2 + W_3 \cdot X_3 + \dots$$

$$= W_0 + \sum_{i=1}^n W_i X_i$$

$$Y = \sum_{i=0}^n W_i X_i = W^T X$$

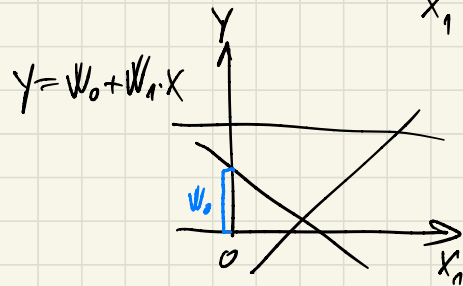
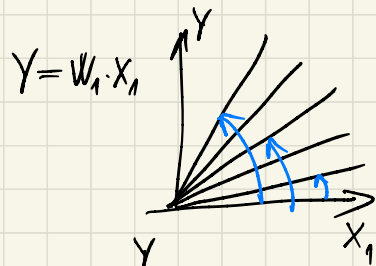
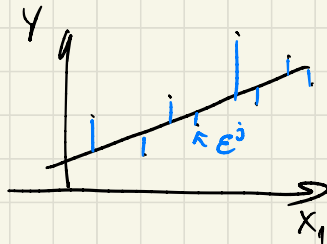
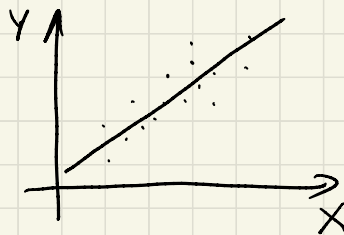
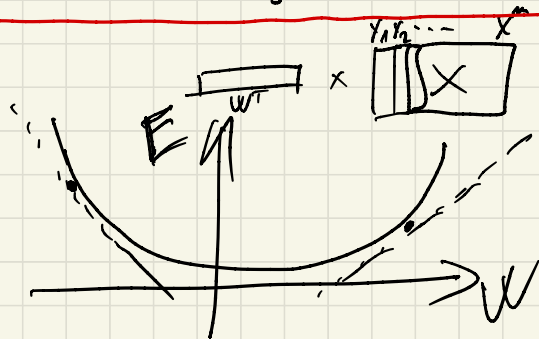
$$X_0 = 1$$

$$W^T \begin{bmatrix} 1 & X_1 & X_2 & X_3 & \dots & X_n \end{bmatrix} \times \begin{bmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \\ \dots \\ X_n \end{bmatrix} =$$

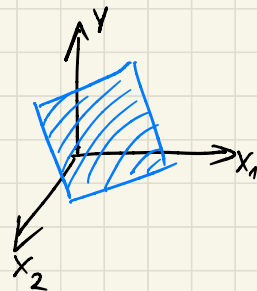
$$Y^j = W^T X^j + \epsilon^j$$

MSE

$$(\min_w) E(W) = \frac{1}{2n} \sum_j (W^T X^j - Y^j)^2 = \frac{1}{2n} \|W^T X - Y\|_2^2$$



$$Y = W_0 + W_1 X_1 + W_2 X_2$$



$$\min E(W) = \frac{1}{2m} \sum_j (W^T X^j - Y^j)^2$$

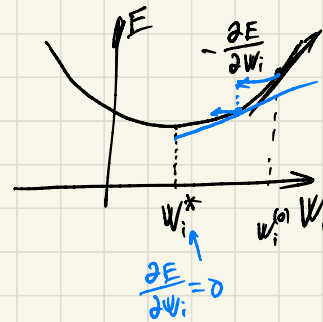
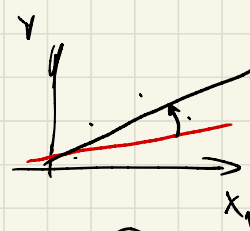
jedna tačka: $X = X^j$

$$\begin{aligned} \frac{\partial E}{\partial W_i} &= \frac{2}{2m} \left[\frac{1}{m} (W^T X - Y)^2 \right] \\ &= \frac{1}{2m} 2 \cdot (W^T X - Y) \cdot \frac{\partial}{\partial W_i} [W^T X - Y] \\ &= \frac{1}{m} (W^T X - Y) \cdot \frac{\partial}{\partial W_i} \left[\sum_{i=0}^n W_i X_i - Y \right] \\ &= \frac{1}{m} \underbrace{(W^T X - Y)}_{f(x)} \cdot X_i \end{aligned}$$

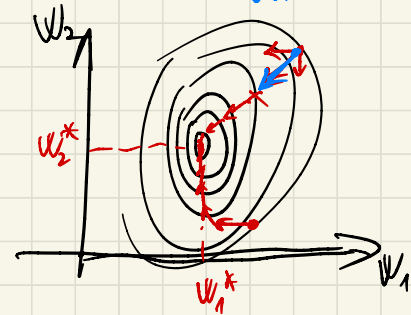
celo X : $[f(x) + g(x)]' = f'(x) + g'(x)$

$$\frac{\partial E}{\partial W_i} = \frac{1}{m} \cdot \sum_j \left[\left(\sum_{i=0}^n W_i X_i^j - Y^j \right) \cdot X_i^j \right]$$

$\forall_i \in [0, n]$



$$\begin{bmatrix} \frac{\partial E}{\partial W_0} \\ \frac{\partial E}{\partial W_1} \\ \frac{\partial E}{\partial W_2} \\ \vdots \end{bmatrix} = \nabla_W E$$



$$W_i^{\text{New}} = W_i^{\text{Old}} - \alpha \frac{\partial E}{\partial W_i}, \forall i$$

ALG: $W^{\text{New}} = W^{\text{Old}} - \alpha \nabla_W E$

α - LEARNING RATE

$$\frac{\partial E}{\partial W_i} = \frac{1}{m} \sum_j \left[\underbrace{\left(\sum_{i=0}^n W_i X_i^j \right)}_{\text{PRED}} - \underbrace{Y_i^j}_{\text{LABEL}} \right] \cdot X_i^j$$

ERROR

$$X = \begin{matrix} & x_1 & x_2 & x_3 & \dots & x_n & x_0 \\ \begin{matrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{matrix} & \begin{matrix} \square \\ \square \\ \square \\ \vdots \\ \square \end{matrix} \end{matrix}$$

MATRICE:

$$\begin{matrix} & n \\ m & \begin{bmatrix} \times \end{bmatrix} \end{matrix} \quad \begin{matrix} 1 \\ \begin{bmatrix} Y \end{bmatrix} \\ m \end{matrix}$$

$$\begin{matrix} 1 \\ \begin{bmatrix} W \end{bmatrix} \\ n \end{matrix}$$

PRED: $\begin{matrix} X & \times & W^T \\ (m, n) & (n \times 1) & \end{matrix} \quad \begin{matrix} 1 \\ \begin{bmatrix} \end{bmatrix} \\ m \end{matrix}$

(m, 1)

ERR: $\begin{matrix} \text{PRED} & - & Y \\ (m, 1) & (m, 1) & \end{matrix} \quad \begin{matrix} 1 \\ \begin{bmatrix} \end{bmatrix} \\ m \end{matrix}$

(m, 1)

GRAD: $\frac{1}{m} \begin{matrix} \text{ERR}^T & \times & X \\ (1, m) & (m, n) & \end{matrix}$

ALG:

DATA: $X_0 = 1$

$W = \text{INIT}()$

FOR ITERATIONS:

$\text{PRED} = X \cdot W^T$

$\text{ERR} = \text{PRED} - Y$

$\text{GRAD} = \text{ERR}^T \times X \cdot \frac{1}{m}$

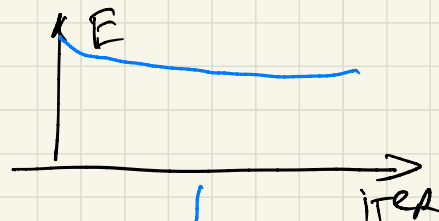
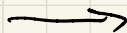
$W = W - \alpha \cdot \text{GRAD}$

$$\begin{matrix} \text{ERR}^T \\ \begin{bmatrix} \end{bmatrix} \end{matrix} \times \begin{bmatrix} X \end{bmatrix} = \begin{matrix} \frac{\partial E}{\partial W_i} & \dots & 1 \\ \begin{bmatrix} \end{bmatrix} & \dots & \end{matrix}$$

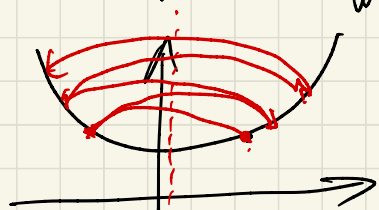
n



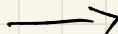
α MALO



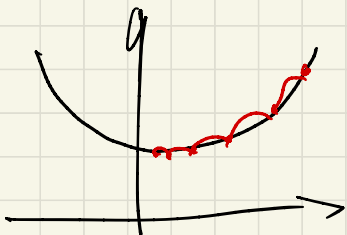
$\Rightarrow \alpha \uparrow$



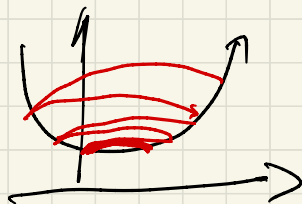
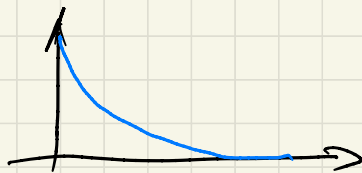
α - JAKO
VELIKO



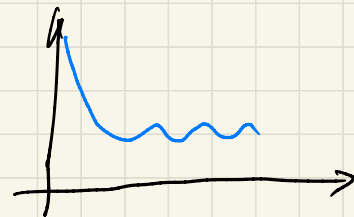
$\Rightarrow \alpha \downarrow$



α OK



α VELIKO



$\alpha \downarrow$

$$E^*(w) = E(w) + \lambda \cdot \sum_{i=1}^n w_i^2$$

↗ ○ PENAL L_2

$$E(w) \rightarrow E^*(w)$$

$$\frac{\partial E^*}{\partial w_i} = ?$$

$E^*(w)$ - convex

GRAD = . . .

λ - SHAGA REGULARIZATION

w_0 - NO REGUL.

L_2 penal $\Rightarrow$ RIDGE

